JLCPCB Impedance Calculator

The JLCPCB Impedance Calculator computes track width values and recommended stack-ups from user-input values of board layer, thickness, copper weight, target impedance, trace spacing (for edge-coupled pairs), and impedance trace to copper gap (coplanar waveguides). Click here to see the user guide of our impedance calculator.

* The first stack-up calculated below is highly recommended as it has lowest cost and quickest turn-around.

Layers	6	PCB Thickness	1.6
Inner Copper Weight	1oz	Outer Copper Weight	1oz
Unit	mil		

Impedance Configure

+ New Impedance | Duplicate Impedance | Calculate

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Spacing (mil)	Impedance trace to copper (mil)

50	Single Ended (Non copl	L1	/	L2	/	/
90	Differential Pair (Non co	L1	/	L2	6	/
100	Differential Pair (Non co	L1	/	L2	7	/

JLC061611-3313D(Special/Finished thickness1.6mm±10%) | JLC061611-3313F(Special/Finished thickness1.57mm±1

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing	Impedance trace to copper
50	Single Ended (Non coplanar)	L1	/	L2	5.6200	/	/
90	Differential Pair (Non coplanar)	L1	/	L2	5.2100	6.0000	/
100	Differential Pair (Non coplanar)	L1	/	L2	4.3000	7.0000	/

Layer	Material	Thickness (mil)	Thickness (mm)
L1	Outer Copper Weight1oz	1.38	0.0350
Prepreg	3313 RC57% 4.2mil	3.62	0.0920
L2	Inner Copper Weight	1.18	0.0300
Core	0.3mm 1/1OZ without copper	11.81	0.3000
L3	Inner Copper Weight	1.18	0.0300
Prepreg	7628, RC 49%, 8.6 mil	7.99	0.2030
Prepreg	7628, RC 49%, 8.6 mil	8.58	0.2180
Prepreg	7628, RC 49%, 8.6 mil	7.99	0.2030

L4	Inner Copper Weight	1.18	0.0300
Core	0.3mm 1/1OZ without copper	11.81	0.3000
L5	Inner Copper Weight	1.18	0.0300
Prepreg	3313 RC57% 4.2mil	3.62	0.0920
L6	Outer Copper Weight1oz	1.38	0.0350

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